COMPREHENSIVE AUDIT & CHANGELOG

WealthFund Calling CRM — Complete Changes & Technical Architecture Report

Detailed breakdown of all features, cPanel cloud adaptations, database optimizations, and system enhancements

PROJECT	GITHUB REPOSITORY	TARGET ENVIRONMENT	GENERATED DATE
Calling CRM (v3.0)	pranav8799/wealthfunds28102026	BigRock cPanel / Cloud Linux	October 28, 2026

1 Executive Summary & Recent cPanel Deployment Changes

This report documents the exhaustive list of modifications made across the **Calling CRM** system, with special focus on transforming the local XAMPP-based installation into an enterprise-ready, cloud-hosted production web application running on **BigRock cPanel**.

> **Key Milestone Achieved:** The CRM was fully ported, bundled, git-versioned, and configured to execute via CloudLinux Phusion Passenger in BigRock cPanel. The application requires zero compilation overhead on shared hosting servers and features built-in database resilience.

File / Component	Change Type	Description & Impact
`server/src/db.js`	FIX DATABASE	**Handled cPanel Database Permission Restrictions:** Rewrote `db.init()`. On cPanel shared hosting, database users are restricted to their assigned database (`user_dbname`) and lack global `CREATE DATABASE` privileges. The code now catches privilege denial gracefully and verifies connection directly via the pool. Added diagnostic alerts for `ER_BAD_DB_ERROR` and `ER_ACCESS_DENIED_ERROR`.
`server/src/index.js`	ENHANCE	**Flexible Static Dist Routing:** Updated Express to dynamically detect the built React frontend from candidate paths (`client/dist`, `server/dist`, `dist`, `public`). Automatically logs the active static directory and serves React SPA routes while preserving `/api/*` endpoints.
`server/app.js` & `app.js`	NEW	**Phusion Passenger Entry Points:** Created entry points at both the server root and project root required by CloudLinux "Setup Node.js App" in cPanel to execute the backend cleanly.
`database/cpanel_import.sql`	NEW	**cPanel-Safe Database Script:** Created a clean SQL import schema without hardcoded `CREATE DATABASE` and `USE calling_crm` commands. Enables error-free direct import via phpMyAdmin in cPanel without `#1044 Access Denied` failures.
`package-cpanel.bat`	NEW	**One-Click Production Packager:** Created a Windows batch automation script that triggers Vite production build, copies frontend assets to `server/dist`, and packages all server files into a lightweight zip (`calling-crm-cpanel.zip`) of only 288 KB (excluding bulky `node_modules`).
`.gitignore`	MOD	**Tracked Prebuilt Frontend in Git:** Modified Git ignore rules to whitelist and track `!server/dist/` so when cPanel clones or pulls from GitHub, the compiled React application is immediately available without running `npm run build` on shared hosting.
GitHub Repository Migration	MIGRATE	**New Remote Origin & Initial Push:** Initialized, linked, and pushed the entire modernized repository to `https://github.com/pranav8799/wealthfunds28102026.git` on branch `main`.

2 Complete Module-by-Module Project Changes & Features

A. Lead Ingestion, Parsing & Google Sheets Synchronization

- **Multi-Format Import Support:** Integrated parser for **Excel (.xlsx, .xls)** and **CSV** files with memory buffering (25MB limit) using `multer` memory storage (no residual files left on disk).
- **Google Sheets Real-Time Sync:** Added URL-based sheet fetcher. When a published Google Sheet is linked, the CRM auto-synchronizes new entries every 15 minutes (`SYNC_MINUTES`) in the background.
- **Intelligent Column Detection:** Automatic header detection and mapping algorithm that identifies client name and telephone numbers across diverse sheet headers.
- **Data Normalization:** Strips extraneous characters, international dialing prefixes (`+91` , `0` , `p:`), and enforces strict 10-digit mobile phone sanitization.
- **Duplicate Protection:** Database-level uniqueness on `leads.mobile` ensures existing clients are never duplicated during mass imports.

B. Automated Lead Distribution Engine (v3 Cutoff Logic)

- **Background Distribution Runner:** Automated daemon running every 60 seconds (`inbox.distributeFresh()`) that balances uncalled leads among active team members.
- **Pre-12:00 PM Distribution:** Prior to 12:00 PM cutoff, leads are distributed equally across all active executives who are not flagged as Absent or on Leave.
- **Post-12:00 PM Real-Time Rebalancing:** After 12:00 PM, executives who have not logged in are identified; their untouched, uncalled fresh leads are automatically reclaimed and redistributed among present, logged-in executives.
- **Late Login Re-allocation:** If an executive signs in after 12:00 PM, the system recognizes their presence and assigns their proportional quota of fresh leads.
- **Sunday / Weekly Off Rule:** Respects configurable weekly off days (`WEEKLY_OFF=0` for Sunday), distributing exclusively to staff who sign in voluntarily.

C. Calling Desk & Two-Way Mobile Device Synchronization

- **PC Calling Desk:** Interactive calling dashboard with live stopwatch timer, client history, previous call remarks, and immediate status updates.
- **Web Phone Mode:** Mobile web application interface (`/executive/phone`). When an executive clicks "Call" on their PC, their connected smartphone screen vibrates and automatically loads the dial pad with the client's number.
- **QR Code Fallback:** Automatic on-screen QR code generation allows instant dialer launch by scanning the screen if web sockets/Wi-Fi are disconnected.
- **Native Android Companion App:** Integrated Android background dialer application (`android-app/`) capable of detecting native Android telephony call state changes and reporting exact talk durations directly to MySQL.
- **Tamper-Proof Duration Metrics:** Call duration is computed by the backend server based on verified call timestamps, preventing executives from manually manipulating talk times.

D. Security, Device Verification & Role-Based Access Control

- **Role Separation:** Segregated routing and permission layers for **Admin** (`/admin/login`) and **Executive** (`/login`).
- **Bcrypt Password Hashing:** Passwords stored using salted `bcryptjs` hashes.
- **Hardware Device Fingerprinting:** System tracks device hashes in `trusted_devices` table. Any login from an unfamiliar PC, phone, or browser requires OTP validation.
- **Dual OTP Modes:** Supports both **SMS Gateway** (Powerstext DLT-approved routes) and **Admin-Assisted OTP** (displayed directly on Admin panel for instant verbal verification without SMS cost).
- **Brute-Force Safeguards:** Enforces 30-second delays between OTP requests, maximum 8 OTP requests per hour, and automatic invalidation after 5 consecutive incorrect attempts.

E. Attendance & Comprehensive Activity Auditing

- **Automated Attendance Tracking:** Executives are automatically registered as "Present" upon their initial sign-in of the day with exact timestamp logging.
- **Admin Attendance Grid:** Full-month calendar view allowing administrators to record leaves, half-days, and absences with custom notes.
- **Tamper-Proof Activity Log:** High-granularity logging of every sign-in, call outcome, lead status transition, reassignment, remark edit, and batch import with executive ID, client ID, client IP, and UTC timestamps.
- **Lead Sharing & Notes:** Inter-executive lead sharing module with transfer notes; transferred leads surface directly in the recipient's "Shared with me" filter tab.

3 Database Schema & Automated Migrations

The database architecture is designed with self-healing tables and automatic version migrations. Existing databases automatically receive missing columns upon server startup without manual SQL execution.

Table Name	Primary Keys & Indexes	Purpose & Key Columns
`users`	`id` (PK), `uq_users_phone`, `uq_users_name`	Stores administrator and executive profiles, password hashes, role, active status, personal Zoom links, and last login time.
`leads`	`id` (PK), `uq_leads_mobile`, `idx_leads_inbox`	Core client repository. Tracks executive assignment, status (NEW, INTERESTED, etc.), remark history, webinar progress, account opening status, import source, and inbox date.
`calls`	`id` (PK), `idx_calls_lead`, `idx_calls_exec`	Call interaction records. Tracks start timestamp, hangup timestamp, total duration in seconds, call outcome, and executive remarks.
`activity_log`	`id` (PK), `idx_act_user`, `idx_act_created`	Audit log recording user actions, IP addresses, user agents, lead edits, client reassignments, imports, and password updates.
`attendance`	`id` (PK), `uq_att(user_id, date)`	Daily work attendance ledger. Stores status (PRESENT, ABSENT, LEAVE, HALF_DAY), first login time, last seen time, and admin remarks.
`trusted_devices`	`id` (PK), `uq_dev(user_id, hash)`	Authorized hardware and browser fingerprint whitelist for bypassing repetitive OTP authentication.
`login_otps`	`id` (PK), `idx_otp_user`	Tracks 6-digit verification codes, expiration timestamps (10-minute TTL), device hashes, and attempt counters.
`lead_shares`	`id` (PK), `idx_share_lead`, `idx_share_to`	Audit history of internal lead transfers between executives, including transfer timestamps and sender handover notes.

Real-Time SQL Triggers

To ensure phpMyAdmin and raw database queries reflect readable data without complex joins, four active database triggers maintain real-time copies of executive names across tables:

- `trg_leads_exec_name_ins`: Automatically sets `leads.executive_name` on INSERT based on `leads.executive_id`.
- `trg_leads_exec_name_upd`: Updates `leads.executive_name` whenever a lead is reassigned to another executive.
- `trg_calls_exec_name_ins`: Automatically records `calls.executive_name` when a new call record is initiated.
- `trg_users_name_upd`: Propagates executive name updates across all historical leads and calls if an executive is renamed in Admin panel.

4 Production Environment Configuration Reference (`server/.env`)

Variable	Recommended Setting	Description
`DB_HOST`	`localhost`	cPanel MySQL host address (use localhost on BigRock servers).
`DB_PORT`	`3306`	Default MySQL port.
`DB_USER`	`cpaneluser_crmuser`	cPanel database user with ALL PRIVILEGES granted on the database.
`DB_PASSWORD`	`********`	Strong password generated via cPanel Password Generator.
`DB_NAME`	`cpaneluser_crm`	Database name created in cPanel MySQL Databases.
`PORT`	`5000`	Application port (Passenger binds automatically via environment socket).
`JWT_SECRET`	`[random 64-char string]`	Cryptographic secret key for signing user authentication tokens.
`APP_TZ`	`Asia/Kolkata`	Application time zone for distribution cutoff and attendance.
`OTP_MODE`	`admin` (or `sms` / `both`)	OTP verification method for newly registered hardware devices.
`DISTRIBUTION_CUTOFF_HOUR`	`12`	Daily hour threshold (12:00 PM) for absent executive lead rebalancing.